

Industrial Security (Bachelor)

Solution Sheet – Section 07

Risk Management

Exercise 7.1 – Risk Matrix (sample)

Sample top-three risks for a gas compressor station:

1. Loss of control of compressor speed (consequence: catastrophic; likelihood: possible) – mitigate via independent overspeed trip and SIS isolation.
2. Ransomware on Historian server (consequence: major; likelihood: likely) – mitigate via immutable backups and segmentation.
3. Spoofed alarms / suppressed alarms (consequence: major; likelihood: possible) – mitigate via independent alarm path, change-control on alarm configuration.

Exercise 7.2 – Cyber-HAZOP (sample row)

Guide word: More **Parameter:** Flow

Cyber cause: Attacker increases pump set-point via HMI compromise.

Safeguards: Alarm at 1.1× design; SIS trip at 1.3× design.

Gap: No Modbus write allowlist on the HMI-PLC conduit.

Exercise 7.3 – CCE (sample)

Worst-day consequence: contamination of supply due to loss of disinfection. Minimum failure set: chlorine dosing pump command + chlorine residual sensor reading. Engineering change: install a mechanical interlock that closes a downstream valve if residual sensor reads below threshold for 30s, independent of the BPCS.

Exercise 7.4 – Risk Response

1. **Avoid.** The cost of decommissioning the fax line is small and removes an attack surface.
2. **Mitigate.** Replace OPC Classic with OPC UA over TLS or use a UA Gateway in the iDMZ.
3. **Mitigate + Transfer.** Reduce to brokered/recorded sessions, contractually require vendor security controls.
4. **Avoid.** Remove Internet exposure immediately; deploy VPN/jump host.